Gauge-Averaged Confinement in Relative SU(2) BFSS: Explicit Sector Thresholds, Gaps, and Perturbation Bounds

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Abstract

We examine the declared closed-form relative SU(2) BFSS operator, without treating finite Ritz levels as positive embedded eigenvalues. Exact double counting of the transverse pair potentials and a sharp color-orbit average strengthen the source's confinement coefficient while preserving its physical gauge law and rotation-sector premise. The resulting two-parameter kinetic/radial bound gives explicit sector thresholds, a positive Hardy-based spectral gap, tail bounds, and a quantitative class of symmetry-preserving form perturbations. The Clifford normalization also gives explicit fermionic constants without constructing the full fermionic matrix. A solvable coupling example shows why the perturbation conclusion cannot be extended to arbitrary small symmetry-breaking errors. The source's form identity, branching and sector-supply mechanism are retained as named inputs; the unrestricted spectral set and global positive-eigenvalue claim are not independently certified by our finite checks. Contribution assessment: C3/5, the gauge-averaged sector threshold and explicit gap and perturbation budgets are notable source-specific refinements, not an independent proof of the global BFSS claim.

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1 The Physical Operator and the Source Mechanism

The assigned manuscript [1] removes the free center of mass and writes the nine traceless matrices in the color basis $T_a = \sigma_a/\sqrt{2}$. Thus

$$\operatorname{tr}(T_a T_b) = \delta_{ab}, \qquad f_{abc} = \sqrt{2}\,\varepsilon_{abc}, \qquad x = (x^1, x^2, x^3) \in (\mathbb{R}^9)^3, \quad r = |x|.$$

The fermionic space $\mathcal{F}$ is the irreducible complex module of 48 real Clifford generators, normalized by $\{\theta^b_\beta, \theta^c_\delta\} = \delta_{bc}\delta_{\beta\delta}$. It has dimension 2^{24}. The real symmetric 16-dimensional gamma matrices satisfy $\{\gamma^i, \gamma^j\} = 2\delta_{ij} I$. Inner products below are conjugate linear in the first argument.

Let $O_h \in \mathrm{SO}(3)$ be the adjoint color action of $h \in \mathrm{SU}(2)$, and V_h its unitary action on $\mathcal{F}$. The physical law is

$$(\mathcal{G}_h \Psi)(x) = V_h \Psi(O_h^{-1} x), \qquad \mathcal{H} = [L^2(\mathbb{R}^{27}; \mathcal{F})]^{\mathrm{SU}(2)}, \qquad \mathcal{C} = [C_c^\infty(\mathbb{R}^{27}; \mathcal{F})]^{\mathrm{SU}(2)}.$$

Lebesgue measure is used on all 27 real coordinates, with the stated fermionic inner product. This is not a quotient coordinate measure with an omitted Jacobian. The source defines the nonnegative form by closing

$$q[\Psi] = \frac{1}{16} \sum_{\alpha=1}^{16} \|Q_\alpha \Psi\|_2^2, \quad Q_\alpha = -i\gamma^i_{\alpha\beta}\theta^a_\beta \partial_{ia} + \tfrac{1}{2} f_{abc} x^b_i x^c_j \gamma^{ij}_{\alpha\beta} \theta^a_\beta.$$

Its associated self-adjoint operator is H. We retain the source core identity

$$q[\Psi] = \tfrac{1}{2} T[\Psi] + \int V(x) \|\Psi(x)\|^2 \, \mathrm{d}x + \int \langle \Psi(x), B(x)\Psi(x) \rangle \, \mathrm{d}x, \tag{1.1}$$

where

$$T[\Psi] = \|\nabla \Psi\|_2^2, \quad V = \sum_{a<b} |x^a \wedge x^b|^2, \quad B = \tfrac{i}{2} f_{abc} x^a_i \gamma^i_{\beta\delta} \theta^b_\beta \theta^c_\delta.$$

All later constants depend on this exact normalization.

1.1 What is inherited, and what is strengthened

The entire 888-line scientific source and its README were freshly read. Its proof has the following steps, which should not be compressed into a finite matrix claim.

1. Lines 113–132 close the supercharge form; lines 236–325 derive (1.1) and a finite linear fermionic bound.

2. Lines 327–411 construct commuting gauge and Spin(9) actions. Character projection P_λ preserves the form core, and the isotypic space $\mathcal{H}_\lambda$ reduces H.

3. Lines 439–558 give the B_4-to-D_4 branching inequality $\nu_1 \geq \lambda_2$ and justify restriction of smooth isotypic vectors to each transverse slice. Evaluation is done on a finite translate span, not on arbitrary L^2 representatives.

4. Lines 560–685 exclude low oscillator levels and prove sector compactness by a gradient bound, a radial tail bound and Rellich.

5. Lines 696–803 supply actual infinite-dimensional physical sectors; lines 805–819 promote their eigenvectors using exact reduction.

The present contribution strengthens step 4, makes constants in step 3 explicit, and gives consequences conditional on these source interfaces. It does not claim that reading the whole manuscript independently certifies every Clifford, representation-theoretic and global spectral assertion. In particular, the source's stated refutation of an original BFSS exclusion is attributed to the source, not established by a Ritz calculation here.

For clarity we use the following retained premises: (1.1) on $\mathcal{C}$; exact reducing isotypic projections with projected form cores; the source's slice-type inequality; and, when asserting a nonempty infinite sequence, the source's physical sector-supply construction. The local estimates below are proved under these precise premises.

2 Exact Pair Counting and the Physical Color Average

Fix a color a, put $u = x^a \neq 0$, and decompose $x^b = s_b \widehat{u} + z_b$, $z_b \perp u$, for $b \neq a$. For fixed u, this is an orthogonal change of the other 18 coordinates; its Jacobian is one. Write

$$V_a(x) = \sum_{b \neq a} |x^a \wedge x^b|^2 = |u|^2 \sum_{b \neq a} |z_b|^2, \qquad T_a[\Psi] = \int \sum_{b \neq a} |\Pi_{(x^a)^\perp} \nabla_b \Psi|^2 \, \mathrm{d}x.$$

The value of the projection on $x^a = 0$ is irrelevant on a null set. The projection acts on the spatial derivative index and componentwise on $\mathcal{F}$. No derivative with respect to the moving parameter u is taken in this definition.

Lemma 2.1 (Exact pair budget)**.** *For smooth compactly supported vectors,*

$$\sum_{a=1}^{3} V_a = 2V, \qquad \sum_{a=1}^{3} T_a \leq 2T.$$

If every transverse oscillator component has level $n \geq n_0$, put $L = n_0 + 8$. For every $\alpha > 0$,

$$\alpha T[\Psi] + \int V \|\Psi\|^2 \geq \sqrt{\alpha} L \int \sum_a |x^a| \, \|\Psi\|^2. \tag{2.1}$$

Proof. Each unordered pair $\{a, b\}$ occurs twice in $\sum_a V_a$. For fixed b, exactly two transverse projections occur and each has norm at most one. This proves both identities/inequalities. The 16-dimensional oscillator $-\alpha\Delta_z + |u|^2|z|^2$ has energies $2\sqrt{\alpha}\,|u|(n+8)$, by dilation of the standard oscillator. Applying its lower bound on each admissible slice and using Fubini gives

$$\alpha T_a[\Psi] + \int V_a \|\Psi\|^2 \geq 2\sqrt{\alpha} L \int |x^a| \, \|\Psi\|^2.$$

Half of the sum over a, followed by $\sum T_a \leq 2T$, gives (2.1). This keeps the actual pair potentials rather than replacing each of them separately by the full V. □

Lemma 2.2 (Sharp color-orbit bounds)**.** *For a real 9-by-3 matrix X, with columns x^a, let ω be probability measure on $\mathbb{S}^2$. Then*

$$\frac{1}{2}\|X\|_{\mathrm{F}} \le A(X) := \int_{\mathbb{S}^2} |Xv| \,\mathrm{d}\omega(v) \le \frac{1}{\sqrt{3}}\|X\|_{\mathrm{F}}. \tag{2.2}$$

The lower constant is attained at nonzero rank-one matrices; the upper constant is attained when $X^T X$ is scalar. For every physical vector for which the radial integral is finite,

$$\int \sum_a |x^a| \|\Psi(x)\|^2 \,\mathrm{d}x = 3 \int A(X) \|\Psi(x)\|^2 \,\mathrm{d}x \ge \tfrac{3}{2} \int r \|\Psi(x)\|^2 \,\mathrm{d}x. \tag{2.3}$$

Proof. After singular-value decomposition and division by $\|X\|_{\mathrm{F}}$, write the squared singular values as $t_i \ge 0$, $\sum t_i = 1$. Concavity gives, pointwise, $\sqrt{\sum_i t_i v_i^2} \ge \sum_i t_i |v_i|$. Each coordinate of uniform $\mathbb{S}^2$ has uniform law on $[-1, 1]$, so its mean absolute value is $1/2$. This proves the lower bound and the rank-one equality. Jensen's inequality and $\int v_i v_j \,\mathrm{d}\omega = \delta_{ij}/3$ give the upper bound, with equality for three equal singular values. The zero matrix is immediate.

Gauge invariance means $\Psi(O_h x) = V_h \Psi(x)$, hence $\|\Psi(O_h x)\|^2 = \|\Psi(x)\|^2$ almost everywhere. Lebesgue measure is color invariant. Haar averaging a fixed column norm therefore replaces it by $A(X)$, since the adjoint map covers SO(3). Tonelli justifies this also before proving finiteness. Summing three columns proves (2.3). The fermionic vector is not held fixed under the gauge action; its unitary norm is what is used. □

Theorem 2.3 (Gauge-averaged confinement)**.** *Assume the retained source premises and a fermionic bound $\|B(x)\|_{\mathrm{op}} \le Cr$. Suppose the source slice selection excludes levels $n < n_0$ in $\mathcal{H}_\lambda$, and put $L = n_0 + 8$. For $0 < \alpha < 1/2$, set*

$$\eta = \tfrac{1}{2} - \alpha, \qquad c = \tfrac{3}{2} L\sqrt{\alpha} - C.$$

If $c > 0$, then on the entire restricted form domain,

$$q[\Psi] \ge \eta \|\nabla \Psi\|_2^2 + c \int r \|\Psi(x)\|^2 \,\mathrm{d}x. \tag{2.4}$$

In particular $\Psi \in H^1$ and $r^{1/2}\Psi \in L^2$. Such an α exists whenever

$$L > \frac{2\sqrt{2}}{3} C. \tag{2.5}$$

Proof. On the core, insert (2.1) and (2.3) into (1.1). This proves (2.4). Approximate a restricted-domain vector by its projected form-core sequence. Applying the inequality to differences makes both gradients and $r^{1/2}$-weighted vectors Cauchy in L^2. Closedness of these two operators and passage to the limit prove the domain assertion and inequality. Finally α must lie strictly between $(2C/(3L))^2$ and $1/2$, which is possible precisely under (2.5). No claim is made at equality by this argument. □

The source uses $n_0 = m + 1$, hence $L = m + 9$. With the same reserved quarter of kinetic energy, the comparison is

estimate	gradient coefficient	radial coefficient
source	1/4	$L/3 - C$
exact pair count only	1/4	$L/2 - C$
pair count and physical gauge average	1/4	$3L/4 - C$.

The orbit constant is sharp as a matrix inequality; the whole operator bound is not asserted sharp. Other inequalities, particularly the fermionic norm and discarded derivative directions, can be strict.

3 Explicit Fermionic Constants

These calculations concern the stated finite Clifford representation, not a numerical diagonalization of its 2^{24}-dimensional space.

Proposition 3.1 (Weight and operator bounds)**.** *In the source normalization one may take*

$$M = 6, \qquad \|B(x)\|_{\mathrm{op}} \le 8\sqrt{3}\, r. \tag{3.1}$$

Here M bounds the first coordinate of every Spin(8) *weight of $\mathcal{F}$; it is the exact maximum on the full $\mathcal{F}$. The norm constant is only an explicit upper bound, not claimed optimal.*

Proof. A rotation through angle t in one spatial two-plane acts on $\mathbb{R}^{16}$ by $\exp(t\gamma^{12}/2)$, up to the harmless orientation sign. The real skew matrix γ^{12} squares to $-I$, so it consists of eight two-plane rotations with angular speed of absolute value one. The spinor action therefore has eight planes with speed $1/2$. On the three color copies there are 24 such planes. Lifting to the Clifford module halves each speed again: the commuting fermionic weights are sums of 24 independent terms $\pm 1/4$. The largest first coordinate is $24/4 = 6$, attained by choosing all positive occupations. Transport to any conjugate stabilizer preserves this bound. It does not assert that the extremal weight occurs in the gauge-singlet fermionic subspace.

Index the 48 generators by $j = (b, \beta)$, and define the real skew matrix

$$A(x)_{b\beta,c\delta} = \sum_{a,i} f_{abc} x_i^a \gamma^i_{\beta\delta}.$$

Then $B = (i/2)\sum_{j,k} A_{jk}\theta_j\theta_k$. Orthogonal skew normal form gives blocks $\left(\begin{smallmatrix} 0 & \kappa_l \\ -\kappa_l & 0 \end{smallmatrix}\right)$, $\kappa_l \ge 0$, $1 \le l \le 24$. In that basis $B = \sum_l i\kappa_l \theta_{2l-1}\theta_{2l}$. The commuting factors have eigenvalues $\pm\kappa_l/2$, and the irreducible Clifford module realizes every occupation. Consequently

$$\|B\|_{\mathrm{op}} = \tfrac{1}{2}\sum_{l=1}^{24} \kappa_l, \qquad \sum_l \kappa_l^2 = \tfrac{1}{2}\|A\|_{\mathrm{F}}^2.$$

The gamma trace identity and the color normalization give

$$\mathrm{tr}(\gamma^i\gamma^j) = 16\delta_{ij}, \qquad \sum_{b,c} f_{abc} f_{a'bc} = 4\delta_{aa'}, \qquad \|A(x)\|_{\mathrm{F}}^2 = 64r^2.$$

Cauchy–Schwarz now yields $\|B\| \le \tfrac{1}{2}\sqrt{24 \cdot 32}\, r = 8\sqrt{3}\, r$. Zero skew blocks are allowed throughout. □

By the source branching premise, any nonzero level must satisfy $\lambda_2 \le n + 6$. Thus the exact integer lower level available from this information is

$$n_0 = \max\{0, \lceil \lambda_2 - 6 \rceil\}, \qquad L = n_0 + 8. \tag{3.2}$$

The ceiling matters for half-integral weights. This is not a proof that every level above n_0 actually occurs.

Corollary 3.2 (A concrete sufficient sector threshold)**.** *Under the retained source premises, any nonzero physical isotypic sector with $\lambda_2 > 11$ obeys*

$$q[\Psi] \ge \frac{1}{18}\|\nabla\Psi\|_2^2 + (L - 8\sqrt{3}) \int r\|\Psi\|^2, \qquad L \ge 14. \tag{3.3}$$

Proof. Use $\alpha = 4/9$, so $\eta = 1/18$ and $(3/2)\sqrt{\alpha} = 1$. Formula (3.2) gives $L \geq 14$, and $14 > 8\sqrt{3}$ follows by squaring positive quantities. □

For a comparison using the same explicit $C = 8\sqrt{3}$ and $M = 6$, the source's displayed choice requires $m + 9 > 24\sqrt{3}$, for which the least nonnegative integer is $m = 33$, and hence $\lambda_2 > 39$. The pair-count-only fixed-quarter estimate allows $m = 19$, $\lambda_2 > 25$; the gauge-averaged fixed-quarter estimate allows $m = 10$, $\lambda_2 > 16$. Allowing the smaller but positive gradient reserve in (3.3) gives $\lambda_2 > 11$. These are sufficient thresholds from particular estimates, not exact phase boundaries or a classification of all confined sectors.

4 Quantitative Gap, Tails, and the Kinetic Split

Proposition 4.1 (A dimension-27 gap certificate)**.** *If* (2.4) *holds with* $\eta, c > 0$*, then*

$$q[\Psi] \geq g(\eta, c)\|\Psi\|_2^2, \qquad g(\eta, c) = 3\left(\frac{625\eta c^2}{16}\right)^{1/3}. \tag{4.1}$$

Moreover, for every $R > 0$*,*

$$\int_{r>R} \|\Psi\|^2 \leq \frac{q[\Psi]}{cR}, \qquad \|\nabla\Psi\|_2^2 \leq \frac{q[\Psi]}{\eta}. \tag{4.2}$$

The restricted resolvent is compact. If the sector is infinite dimensional, its eigenvalues tend to infinity and are all at least $g(\eta, c) > 0$*.*

Proof. The classical Hardy inequality in dimension $d = 27$ is

$$\int |\nabla\Psi|^2 \geq \frac{(d-2)^2}{4} \int r^{-2}\|\Psi\|^2.$$

For smooth vectors away from zero it follows by expanding $\int |\nabla\Psi + \frac{d-2}{2}\frac{x}{r^2}\Psi|^2 \geq 0$ and integrating the cross term using $\operatorname{div}(x/r^2) = (d-2)/r^2$. Approximation extends it componentwise to H^1; no boundary trace at the origin is imposed. Put $A = 625\eta/4$. The pointwise minimum of $A/r^2 + cr$, $r > 0$, occurs at $r = (2A/c)^{1/3}$ and is $3(Ac^2/4)^{1/3}$, proving (4.1). The two bounds in (4.2) follow directly from the form bound.

A form-norm bounded sequence is bounded in H^1, hence has locally L^2-convergent subsequences by Rellich, componentwise in finite $\mathcal{F}$. The uniform tail estimate makes a diagonal subsequence globally Cauchy in L^2. This proves compactness of the form-domain inclusion and therefore of the restricted resolvent. The remaining claims follow from the compact-resolvent spectral theorem. Promotion of these eigenvectors to the full operator still uses exact reduction. □

The gap is explicit but conservative: Hardy's inequality and the pointwise radial minimum need not be simultaneously attained by a sector vector. For the first guaranteed level $L = 14$, (3.3) gives

$$g_0 = 3\left(\frac{625(14 - 8\sqrt{3})^2}{288}\right)^{1/3} > 0. \tag{4.3}$$

For example, a normalized eigenvector with eigenvalue E has outside-ball mass at most $E/[(14 - 8\sqrt{3})R]$. This is an energy dependent radius, not localization at a single fixed radius for the entire unbounded eigenvalue sequence.

Proposition 4.2 (Best split within this gap bound)**.** *Set $D = 3L/2$, and assume $D > \sqrt{2}C$. Among admissible splits $0 < \alpha < 1/2$, $D\sqrt{\alpha} > C$, the largest lower bound in (4.1) is obtained at*

$$\sqrt{\alpha_*} = \frac{C + \sqrt{C^2 + 4D^2}}{4D}. \tag{4.4}$$

This optimizes this certificate only, not the exact spectral gap.

Proof. The cube of the bound is a fixed positive multiple of $F(t) = (1/2 - t^2)(Dt - C)^2$, where $C/D < t < 1/\sqrt{2}$. Differentiation gives

$$F'(t) = 2(Dt - C)(D/2 - 2Dt^2 + Ct).$$

The second factor has exactly one positive root, given by (4.4); it lies strictly in the admissible interval when $D > \sqrt{2}C$. Since F vanishes at both endpoints and is positive inside, this is its unique maximum. □

4.1 Gauge-compatible finite truncation

A sharp cutoff to a ball is generally not in the form domain. Choose instead a real radial smooth partition $\chi_R^2 + \zeta_R^2 = 1$, with $\chi_R = 1$ on $r \le R$, $\chi_R = 0$ on $r \ge 2R$, and $|\nabla\chi_R|^2 + |\nabla\zeta_R|^2 \le K/R^2$. Such a partition follows by taking cosine and sine of a smooth radial angle. Both functions preserve gauge and rotation sectors. On the core, multiplication terms cancel and the kinetic product rule gives the exact localization identity

$$q[\chi_R\Psi] + q[\zeta_R\Psi] = q[\Psi] + \tfrac{1}{2}\int (|\nabla\chi_R|^2 + |\nabla\zeta_R|^2)\|\Psi\|^2. \tag{4.5}$$

It extends by form approximation. Therefore localization costs at most $K\|\Psi\|^2/(2R^2)$ in total energy, and $\|(1 - \chi_R)\Psi\|^2 \le q[\Psi]/(cR)$. These are compatible analytic truncation bounds. They do not give a complete finite discretization error or an operator residual for a trial vector merely known to be in the form domain.

5 Perturbations That Preserve the Sector

Theorem 5.1 (Explicit admissible perturbation region)**.** *Fix a nonzero source sector satisfying (2.4). Let w be a symmetric sesquilinear form on its form domain such that*

$$|w[\Psi]| \le a\|\nabla\Psi\|^2 + b\int r\|\Psi\|^2 + e\|\Psi\|^2, \quad 0 \le a < \eta, \quad 0 \le b < c, \quad e \ge 0. \tag{5.1}$$

Then $q_w = q + w$ is a closed lower-semibounded form on the same domain, with compact restricted resolvent and lower bound

$$q_w[\Psi] \ge \big[g(\eta - a, c - b) - e\big]\|\Psi\|^2. \tag{5.2}$$

If this number is positive, the whole restricted spectrum is positive. If the sector is infinite dimensional, infinitely many positive eigenvalues tending to infinity remain even without that last condition. To promote them to a perturbed full physical Hamiltonian, it is additionally necessary that a declared full closed form restrict to q_w and that the sector reduce its associated operator.

Proof. Set $\theta = \max(a/\eta, b/c) < 1$. Inequality (2.4) gives $|w[\Psi]| \le \theta q[\Psi] + e\|\Psi\|^2$. Thus the norms defined by $q + \|\cdot\|^2$ and $q_w + (e+1)\|\cdot\|^2$ are equivalent, with positive lower factor $1-\theta$. The latter is complete and defines the claimed closed form; polarization gives its sesquilinear realization. Moreover

$$q_w[\Psi] \ge (\eta - a)T[\Psi] + (c-b)\int r\|\Psi\|^2 - e\|\Psi\|^2.$$

Apply Proposition 4.1, or its Hardy proof, to the positive part. Equivalent form norms preserve compactness of the inclusion. A lower-semibounded compact-resolvent operator on an infinite-dimensional space has eigenvalues tending to infinity, so only finitely many can lie below a fixed positive number. The last sentence of the theorem is the domain and reduction requirement, not a consequence of the norm estimate for w. □

A concrete permitted perturbation is a Hermitian multiplication field $W(x)$ with $\|W(x)\|_{\mathrm{op}} \le br + e$, provided its full form is defined and gauge/rotation covariance actually holds. For example, a scalar radial potential obeys these symmetries. Coefficient changes in the kinetic form may instead contribute the aT term, but the metric, domain and covariance must still be checked. No invariance is inferred from entrywise smallness of a finite matrix.

For a normalized vector with $q_w[\Psi] \le E$, the same proof gives

$$\int_{r>R} \|\Psi\|^2 \le \frac{E+e}{(c-b)R}.$$

The quantitative margin explicitly collapses when $a \uparrow \eta$ or $b \uparrow c$. This is a limitation of the available certificate, not proof of physical delocalization at those boundaries.

6 Why Ritz Levels and Broken Symmetry Are Different

The freshly read original Ram chapters [2, 3, 4] make two relevant distinctions: the Gram matrix carries the physical norm, and the omitted component of the full residual is independent of the finite eigenproblem. In the present sector let $J : \mathbb{C}^k \to \mathrm{Dom}(q) \cap \mathcal{H}_\lambda$ be injective. Its physical mass and form matrices are

$$G_{ij} = \langle \psi_i, \psi_j \rangle, \qquad K_{ij} = q(\psi_i, \psi_j).$$

The generalized problem $Kc = \rho Gc$ is a Ritz problem in the declared form domain. Once compactness is independently established, min–max gives upper estimates for the ordered sector eigenvalues. It does not say that any individual Ritz value is an exact eigenvalue.

If the columns lie in the stronger operator domain, let $P = JG^{-1}J^*$. Then the classical physical residual identity is

$$\|(H_\lambda - z)Jc\|^2 = \|G^{-1}Kc - zc\|_G^2 + \|(I-P)H_\lambda Jc\|^2. \tag{6.1}$$

This follows by orthogonal projection; the first term is not the full residual. For a unit trial vector with a verified full residual at most δ, the self-adjoint spectral theorem gives distance at most δ to the sector spectrum. Compactness then makes the nearby spectral value an eigenvalue. Its promotion to the full operator uses exact sector reduction, not the finite matrix. None of these classical identities is counted as a new M548 result.

Example 6.1 (Arbitrarily small coupling can remove positive point spectrum). Let $\mathcal{K} = L^2((0,\infty), \mathrm{d}t) \oplus \mathbb{C}$, fix $E > 0$, and define

$$H_0(f,c) = (tf, Ec), \qquad W_\epsilon(f,c) = (\epsilon c g, \epsilon\langle g, f\rangle), \qquad g(t) = e^{-t}.$$

Use the multiplication-operator domain for the first component. The perturbation is bounded self-adjoint, with norm $|\epsilon|\|g\| = |\epsilon|/\sqrt{2}$. The unperturbed one-dimensional summand is reducing and has the positive eigenvalue E, inside the spectral set of the other summand. For every real $\epsilon \neq 0$, $H_0 + W_\epsilon$ has no positive eigenvalue. Indeed an eigenvector at $\lambda > 0$ satisfies

$$(t - \lambda)f(t) = -\epsilon c g(t).$$

If $c \neq 0$, the resulting function is not in L^2 near $t = \lambda$, because g is continuous and strictly positive there. If $c = 0$, multiplication by $t - \lambda$ has trivial L^2 kernel, so $f = 0$ also. This is an exact auxiliary model, not a BFSS perturbation claim. It proves that small operator norm alone cannot justify a general persistence assertion for embedded sector eigenvalues after the reducing symmetry is broken.

7 Source Comparison, Costs, and Further Work

7.1 What the estimate does and does not change

The source already supplies a full analytic confinement mechanism, not merely finite Ritz data. Its fixed-color slicing, stabilizer selection, physical sector construction and exact reduction are retained. The new part is an improved use of those same slices: exact pair multiplicity, the physical color-orbit average, explicit Clifford bounds, and the resulting quantitative gap/perturbation region. There is no identified source error here, and no historical priority claim for the classical ingredients.

The source's nonempty sector supply is especially important. It chooses a highest vector v of a rotation type μ in $\mathcal{F}^{\mathrm{SU}(2)}$ and uses the gauge-invariant polynomial

$$P(x) = \sum_{a<b} (\zeta_a \eta_b - \zeta_b \eta_a)^2, \quad \zeta_a = (e_1 + ie_2) \cdot x^a, \quad \eta_a = (e_3 + ie_4) \cdot x^a.$$

The dot products are complex bilinear, not Hermitian. The source proves that $\chi(r)P(x)^k v$ has highest weight $(2k+\mu_1, 2k+\mu_2, \mu_3, \mu_4)$, and disjoint radial supports give infinitely many independent vectors in that same sector. Our threshold applies to sectors supplied by this mechanism but does not infer nonemptiness from a weight inequality alone. Neither μ nor exact eigenvalues are computed in this report.

The benefit is analytic rather than a measured computational speedup. At fixed kinetic reserve the radial coefficient is larger, so the certified tail radius for a given energy is smaller. The explicit constants avoid a search through the entire Clifford matrix. The cost is continued dependence on exact symmetry, full-space form estimates and the original sector-selection proof. The Hardy gap ignores much of the representation and radial structure, so it is not expected to be the best possible gap. Near the sufficient threshold its constants can be poor.

Useful future refinements include optimizing the fermionic skew-matrix nuclear norm using its special gamma/color structure, retaining the actual singular values in $A(X)$, and proving sector-aware radial comparison estimates sharper than Hardy. A validated numerical program would need physical trial functions in the correct domain, a full complement/residual bound, and an analytic infinite-dimensional tail. It would not be justified by increasing the size of a finite matrix alone. A study of symmetry-breaking resonances would be a different problem and is not covered by Theorem 5.1.

7.2 Primary public context checked on October 9, 2026

The primary BFSS abstract [5] proposes a large-N Matrix Theory equivalence. That conjecture is distinct from a statement about the point spectrum of this fixed relative SU(2) realization.

Danielsson–Ferretti–Sundborg [6] report evidence for non-BPS states in a D-particle approximation; this is not a replacement for the exact sector argument. Sethi–Stern [7] address normalizable ground-state invariance, not exclusion of all positive-energy states.

Aref'eva–Koshelev–Medvedev [8] analyze symmetry sectors in an $SU(2) \times SO(2)$ matrix model. The spatial symmetry and model differ from the present Spin(9) problem. Bisi–Zygouras [9] provide classical-group character/branching context; our use of the particular interlacing inequality is through the source's explicit proof. A current 2026 preprint of Ydri [10] concerns mass-deformed Gaussian reduction and singlet counting. Its assumptions do not certify this massless supersymmetric spectral statement. Access to these external papers was limited to primary abstracts and version/publication metadata; their full proofs were not independently audited. Public search results are not a proof of absence of other work.

8 Verification and Scope

Fresh literal reading covered source lines 1–888 in four bounded displays and all 18 README lines. The three original private chapters were read in full: physical metrics, 210 lines; state/domain, 164 lines; and spectral limits, 242 lines. Thus 906 public and 616 private lines were read. Included figures, cited chapters and the rest of the private corpus were not silently counted. Pre-read hashes were persisted before the scientific displays, and final receipts supply posthashes and exact intervals. Complete literal reading is not a claim of independent verification of every imported theorem.

The owned executable checks target pair counting and transverse projections, normalized finite Clifford algebra, the color-average concavity step, split/gap identities and sufficient thresholds. They use exact arithmetic: 242 parameter cases and 723 assertions passed, with no floating-point cases and no parent cases counted. They are local fixtures, not a numerical construction of the claimed BFSS eigenvectors or a proof of sector completeness. Parent checks and prior reports are not counted as owned checks. The full proofs above, rather than sampled matrices, justify the conditional local refinements.

9 Conclusion

The most useful extra information comes from the actual physical law: gauge invariance makes a color-orbit average available, and exact pair counting avoids paying three times for the same potential. Together they improve a source-specific analytic threshold while preserving the domain, measure and symmetry required for spectral promotion. The positive-sequence conclusion remains conditional on genuine reducing, infinite-dimensional sectors and the stated form identities. Applicability L3 and demonstrated local contribution C3 answer different questions.

References

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